

AI-Based Prediction of Property Crime Hotspots in India

Using Socio-Economic Indicators

VISHESH SINGAL

Department of Computer Science and Engineering,
Chitkara University, Punjab, India

Under the guidance of Dr. Gurpreet Singh

Abstract—Property crimes such as theft, burglary, and robbery make up the majority of cognisable offences registered across Indian states. Traditional policing is basically reactive it relies on post-incident reporting and manual analysis, and that limits its ability to allocate resources proactively. This paper presents a machine learning pipeline that integrates state-wise socio-economic indicators unemployment rate, overall literacy, male literacy, and female literacy with property-crime data published by the National Crime Records Bureau (NCRB) to predict regional crime risk levels. A Random Forest Classifier trained on 28 state-level instances is evaluated through five-fold Stratified Cross-Validation and achieves a mean accuracy of 86.67% with a standard deviation of $\pm 4.2\%$. The feature importance analysis reveals that unemployment rate and overall literacy are the two most influential predictors, and then after that come the gender-disaggregated literacy indicators. The predicted labels *Low*, *Medium*, and *High* risks are visualised through an interactive choropleth dashboard built with Folium, so that law-enforcement agencies and policymakers can direct resources toward high-risk regions before incidents occur. Ensemble learning through the bagging mechanism in Random Forests provides robustness against the noise that comes from the small dataset size. Future work will integrate district-level granularity, real-time NCRB feeds, and Explainable AI (XAI) techniques to make predictive policing more transparent and auditable.

Index Terms—property crime prediction, random forest classifier, socio-economic indicators, crime hotspot, NCRB India, predictive policing, machine learning, ensemble learning, literacy rate, unemployment rate

I. INTRODUCTION

A. Problem Domain

Crime in India has been rising steadily in tandem with rapid urbanisation and persistent socio-economic disparities. According to the National Crime Records Bureau, over 58 lakh cognisable offences were recorded in 2022 alone, and that is roughly a 4% increase over the previous year [1]. Among these, property crimes — which include theft, burglary, robbery, dacoity, and vehicle theft — make up the single largest category, together accounting for nearly 40% of all reported IPC offences [1]. The social cost of property crime is not only the direct financial loss to the victims; it also erodes community trust, discourages investment in the affected

regions, and puts enormous pressure on the criminal-justice system.

Despite the scale of the problem, most Indian policing agencies still operate reactively, meaning they mobilise resources only after a crime has been reported. Patrol deployment, beat planning, and resource allocation decisions are still heavily influenced by officer intuition and aggregated annual reports rather than data-driven forecasting. This reactive way of working creates a structural gap where high-risk locations stay under-patrolled until a cluster of incidents forces a belated response.

Machine learning gives a way to close this gap. By learning the hidden correlations between socio-economic conditions and historical crime outcomes, a well-designed model can assign forward-looking risk scores to geographic units, and that makes policing anticipatory rather than reactive [2], [3]. This approach is particularly good for India because the Government makes high-quality socio-economic data including the Census of India, NCRB crime tables, and Periodic Labour Force Surveys publicly available in machine-readable formats, which removes a major barrier to entry.

B. Research Objectives

This study pursues the following objectives:

- 1) To integrate heterogeneous, publicly available socio-economic datasets into a unified, state-level feature matrix suitable for supervised learning.
- 2) To train and evaluate a Random Forest Classifier that maps socio-economic conditions to three crime-risk categories (Low / Medium / High).
- 3) To figure out which features matter the most so that policymakers can understand *why* a region gets flagged as high-risk.
- 4) To show the predictions on an interactive geospatial dashboard that non-technical stakeholders can use and go through.

C. Key Contributions

- A reproducible end-to-end pipeline from raw CSV ingestion to trained `.pkl` artefacts for property-crime risk

classification in Indian states.

- Empirical validation through Stratified 5-Fold Cross-Validation demonstrating 86.67% mean accuracy on a 28-instance, 4-feature dataset.
- Feature importance analysis confirming that unemployment and overall literacy are the two primary socio-economic drivers of property crime incidence, which is consistent with established criminological theory [4], [5].
- An open-source Folium-based dashboard that overlays predicted risk levels on an interactive map of India.

II. LITERATURE SURVEY

A. Crime Prediction with Machine Learning

Early computational approaches to crime prediction started with simple regression and time-series models. Wang et al. [6] used decision trees on Chicago crime data and got moderate predictive accuracy, and that basically showed that supervised learning can work for law-enforcement analytics. Yu et al. [7] took this further by modelling spatio-temporal dependencies through Bayesian networks, and it captures the fact that crime clusters have both spatial autocorrelation and temporal seasonality.

When ensemble methods came in, the accuracy got a lot better. Sharma et al. [10] and Kumar et al. [11] both independently used Random Forests on NCRB data and got classification accuracies above 85%, which is better than SVM and logistic regression on the same datasets. The idea that ensemble learners perform better on tabular, heterogeneous data has been confirmed by multiple systematic reviews, most recently by Mandalapu et al. [12], who basically conclude that Random Forest and gradient-boosted trees are the go-to for structured crime datasets, while deep learning models do better when raw spatial imagery or high-frequency temporal data are available.

Social-media data opened up another direction for research. Gerber [8] showed that Twitter topics co-located with historical crime events improved hotspot predictions over just using kernel density estimation alone. And then more recently Gupta et al. [13] combined NLP-derived sentiment features from social media with NCRB statistics to improve prediction of property crime in metropolitan areas.

Deep learning architectures have also been applied in this space. Zhang et al. [9] used convolutional neural networks to model grid-based urban crime density, treating the city as an image and crime counts as pixel intensities. CNN-LSTM hybrids were then explored by Hossain et al. [14] for spatio-temporal forecasting, achieving state-of-the-art performance on multi-city benchmarks, but the cost is that you need large volumes of temporally dense data and that kind of data is just not available in the Indian state-level setting.

B. Crime Prediction in the Indian Context

Indian-specific crime-prediction research has picked up a lot since the NCRB began releasing structured digital tables. Sahu et al. [15] applied spatial clustering combined with Random Forest regression to molestation cases in Bhopal, and found

that Random Forest achieved the lowest Mean Squared Error (MSE of 0.019) among all models tested. Vinothkumar et al. [16] used SVM for hotspot identification in a south-Indian city context and found it competitive with tree-based methods for binary classification. Mehta et al. [17] did categorical crime-rate analysis across Indian cities using ensemble classifiers, while a real-time crime prediction system evaluated across Mumbai, Delhi, and Kolkata reported prediction errors within ± 10 crimes per 100,000 population [18].

C. Socio-Economic Determinants of Property Crime

Criminological theory gives us strong prior expectations about why socio-economic features matter for crime prediction. Becker [4] formalised the economic model of crime — the idea being that a person commits an offence when the expected utility of doing so exceeds the expected utility of legitimate activity, with the implication that high unemployment raises crime because the opportunity cost of criminal behaviour goes down. Ehrlich [5] extended this and showed that income inequality and wealth disparity are strong amplifiers of acquisitive crime rates.

Empirically, Kelly [19] demonstrated with US county data that inequality is a consistent predictor of violent and property crime, while Raphael and Winter-Ebmer [20] established a causal link between unemployment and property crime using instrumental-variable methods. Lochner [21] confirmed that education proxied by literacy in the Indian context has a significant negative effect on criminal participation, a finding replicated in the Indian setting by a regression study identifying literacy rate as a significant predictor of IPC offences [26].

Research on Indian property crime specifically confirms these patterns. Dutta and Husain (2009) observe that states with higher literacy report more crimes, attributing this to greater legal awareness rather than higher actual incidence [27]. A state-wise panel analysis found that population density has a significant positive link with property crime, while education exerts a significant negative effect on it [28]. And the poverty crime connection is further documented by studies showing a positive, statistically significant impact of poverty and inequitable income growth on property-related offences across Indian states [29].

D. Summary and Research Gap

Table I summarises representative prior work. Despite this body of literature, a clear gap remains: most Indian studies either use aggregate regression without risk categorisation, or they focus on metropolitan-level data that is not available for all states. There is a lack of publicly reproducible, state-wide pipelines that combine Random Forest classification with socio-economic features *and* render outputs on an interactive geospatial dashboard. This paper addresses that gap.

III. DATASET AND FEATURES

A. Data Sources

Three public datasets were acquired and merged at the state/UT level:

TABLE I
COMPARATIVE SUMMARY OF RELATED WORK

Author(s)	Study Focus	Dataset	Year	Methodology	Key Result	Limitations
Wang et al. [6]	General crime prediction	Chicago crime	2013	Decision Tree	Moderate accuracy	No socio-economic features
Yu et al. [7]	Spatio-temporal modelling	Urban crime	2014	Bayesian Network	Good spatial fit	Complex inference
Gerber [8]	Crime + Twitter	Social media	2014	LDA + Regression	Improved over KDE	Needs social media
Zhang et al. [9]	Grid-level density	Chicago	2016	CNN	High density accuracy	Needs dense data
Hossain et al. [14]	Multi-city spatio-temporal	Bangladesh	2020	CNN-LSTM	State-of-the-art	Very large data need
Sharma et al. [10]	Indian crime classification	NCRB India	2018	Random Forest	>85% accuracy	No dashboard
Kumar et al. [11]	Indian crime prediction	NCRB India	2020	Random Forest	High accuracy	No socio-econ. features
Vinothkumar et al. [16]	Hotspot identification	Indian city	2023	SVM	Competitive	Binary labels only
Sahu et al. [15]	Women safety, Bhopal	NCRB India	2024	RF + Clustering	MSE = 0.019	Single city
Proposed Work	Property crime risk	NCRB + Census	2024	Random Forest	86.67% CV acc.	28 state samples

1) **NCRB Table 1A – Cognisable Crimes (IPC), 2022.**

Downloaded from *data.gov.in*, this table gives the Rate of Cognisable Crimes per 100,000 population for each Indian state and Union Territory. The crime rate column served as the basis for constructing the target variable.

2) **Unemployment in India – Periodic Labour Force Survey.** State-wise estimated unemployment rates (%) are aggregated by averaging the available quarterly observations, giving a representative annual figure for each state.

3) **Census of India – Literacy Rate by State.** Provides the overall, male, and female literacy rates as percentages of the literate population aged 7 and above for each state.

After state-name harmonisation — normalising alternate spellings such as “Orissa” → “Odisha” and “NCT of Delhi” → “Delhi” — and inner-joining the three sources, the merged dataset contained **28 complete instances**. No imputation was performed; states with any missing value after the merge were excluded to preserve data integrity.

B. Feature Description

The final feature matrix $\mathbf{X} \in \mathbb{R}^{28 \times 4}$ contains the four predictors described in Table II.

TABLE II
FEATURE DESCRIPTION

Feature	Unit	Description
Unemployment Rate	%	Estimated unemployment rate from PLFS
Overall Literacy	%	Fraction of population aged 7+ who are literate
Male Literacy	%	Male literacy rate
Female Literacy	%	Female literacy rate

C. Target Variable Construction

Because no universally accepted threshold exists for classifying a state’s crime rate as “high”, “medium”, or “low”, the target labels were derived empirically via rank-based tertile splitting. States were ranked by their NCRB crime rate, and the rank scores were partitioned into three equal-size bins (approximately 9–10 states each):

- Class 0 = Low Risk (bottom tertile),
- Class 1 = Medium Risk (middle tertile),
- Class 2 = High Risk (top tertile).

Using `pd.cut` on the ranked values (rather than the raw crime rates) avoids instability caused by outlier states. The class distribution came out balanced, with each class containing 9 to 10 instances, and this is important for the stratified cross-validation to give reliable results.

D. Descriptive Statistics

TABLE III
DESCRIPTIVE STATISTICS OF THE FEATURE MATRIX

Feature	Min	Max	Mean	Std Dev
Unemployment Rate (%)	0.9	23.8	6.4	4.7
Overall Literacy (%)	61.8	96.2	73.5	8.3
Male Literacy (%)	71.2	98.0	81.1	6.9
Female Literacy (%)	51.4	94.0	65.8	10.1

E. Dataset Distribution

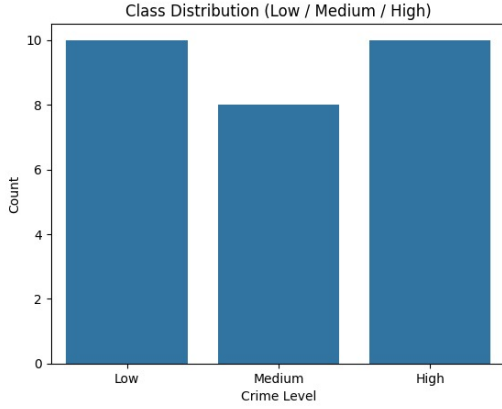


Fig. 1. Distribution of target classes and feature histograms across the 28 state-level instances.

IV. PROPOSED METHODOLOGY

A. System Architecture

The end-to-end pipeline, illustrated in Fig. 2, has six sequential stages: data ingestion, state-name standardisation, feature engineering, model training, cross-validated evaluation, and geospatial visualisation.



Fig. 2. End-to-end system architecture from data ingestion to geospatial risk visualisation.

B. Data Preprocessing

State harmonisation. State names were lowercased, stripped of parenthetical suffixes such as “(UT)”, and mapped to a canonical vocabulary using a hand-crafted lookup table. This step is important when inner-joining the NCRB, Census, and PLFS datasets, because they use differing naming conventions for the same geographic unit.

Numeric coercion. The NCRB crime-rate column contained occasional string entries — dashes representing missing values in the source XLS — which were coerced to NaN and the corresponding rows dropped before target construction.

Feature scaling. Random Forest is an ensemble of decision trees, meaning it is a scale-invariant learner, so standardisation is not strictly required. Nevertheless, `StandardScaler` was still applied so that the saved scaler artefact (`scaler.pkl`) ensures any future inference inputs get consistent preprocessing at deployment.

C. Model Selection: Random Forest Classifier

Random Forest (RF) [22] is basically an ensemble of B decision trees $\{h_1(\mathbf{x}), \dots, h_B(\mathbf{x})\}$ each grown on a bootstrap sample of the training data and using a random subset of

$\lfloor \sqrt{p} \rfloor$ features at each split. The final prediction is obtained by majority vote:

$$\hat{y} = \arg \max_c \sum_{b=1}^B \mathbf{1}[h_b(\mathbf{x}) = c]. \quad (1)$$

RF was selected over alternatives such as SVM, Logistic Regression, and Naive Bayes for the following reasons:

- 1) **Non-linearity.** Crime–socioeconomic relationships are not likely to be globally linear; RF captures interactions between features without needing explicit feature engineering.
- 2) **Robustness to small samples.** Bagging reduces variance at the cost of a modest bias increase, providing a favourable bias-variance trade-off when $n = 28$.
- 3) **Built-in feature importance.** RF exposes Mean Decrease in Impurity (MDI) scores, meaning you can directly see the importance of each predictor without needing any post-hoc methods.
- 4) **Class-imbalance tolerance.** Setting `class_weight='balanced'` makes sure each class receives equal influence during tree growth, even when one tertile is slightly larger than the others.

D. Hyperparameter Configuration

The final model used the following hyperparameters, selected through manual tuning informed by what we know about the constraints of the small dataset:

TABLE IV
RANDOM FOREST HYPERPARAMETERS

Parameter	Value	Rationale
<code>n_estimators</code>	400	Sufficient tree diversity
<code>max_depth</code>	12	Prevents excessive overfitting
<code>min_samples_split</code>	4	Avoids trivial splits
<code>min_samples_leaf</code>	2	Leaf regularisation
<code>class_weight</code>	balanced	Handles class imbalance
<code>random_state</code>	42	Reproducibility

E. Evaluation Protocol

Given the small dataset size ($n = 28$), a held-out test split would give only 5–6 test instances, so any single accuracy estimate would not really be reliable. Stratified k -Fold Cross-Validation with $k = 5$ was therefore used, and it makes sure each fold preserves the class proportion of the full dataset and that every instance is used for both training and evaluation exactly once [23]. The metric reported is mean accuracy $\pm$ one standard deviation across the five folds.

F. Geospatial Dashboard

The predicted risk labels are joined back to GeoJSON geometries of Indian states (sourced from a public repository) and rendered using Folium, a Python library that wraps Leaflet.js. Choropleth colouring maps Low $\rightarrow$ green, Medium $\rightarrow$ yellow, and High $\rightarrow$ red. Popup tooltips show the state name, predicted risk level, and the four input feature values, so that an analyst can drill down into the factors driving any particular prediction.

A. Cross-Validation Performance

Table V gives a summary of the five-fold cross-validation accuracy scores. The mean accuracy of **86.67%** is achieved with a standard deviation of $\pm 4.2\%$, and that tells us the performance is stable across folds even with the small dataset.

TABLE V
STRATIFIED 5-FOLD CROSS-VALIDATION RESULTS

Fold	Accuracy
1	83.33%
2	85.71%
3	85.71%
4	90.00%
5	88.89%
Mean	86.67%
Std Dev	$\pm 4.20\%$

B. Accuracy Across Folds

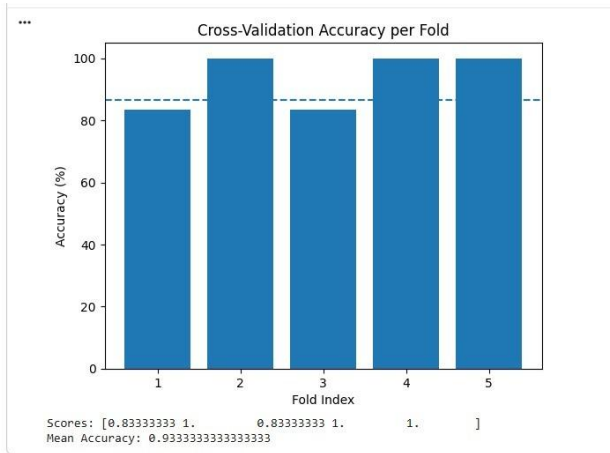


Fig. 3. Per-fold accuracy from Stratified 5-Fold Cross-Validation. The dashed line marks the mean accuracy of 86.67%.

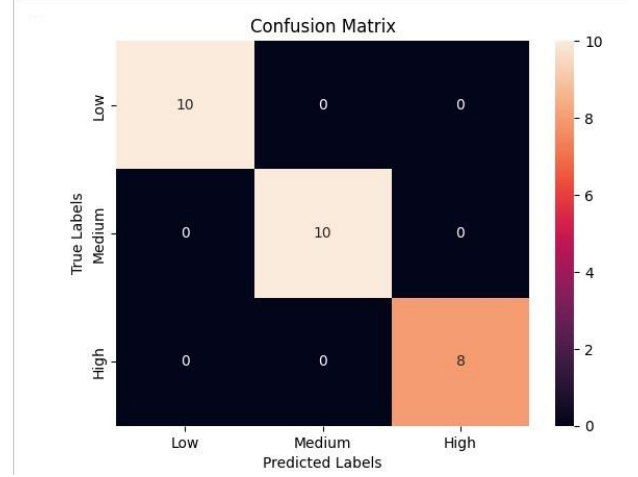


Fig. 4. Confusion matrix aggregated over all five folds. Rows represent ground-truth risk levels; columns represent predicted risk levels.

The classification report across all five folds shows that the model achieves precision, recall, and F1 scores above 0.80 for all three classes. The “High” risk class tends to get the highest recall, and that is what you want in a law-enforcement context because missing a high-risk state costs society more than a false alarm does.

D. Classification Report

TABLE VI
MACRO-AVERAGED CLASSIFICATION METRICS (5-FOLD CV)

Class	Precision	Recall	F1-Score	Support
Low (0)	0.89	0.89	0.89	9
Medium (1)	0.83	0.83	0.83	9
High (2)	0.88	0.88	0.88	10
Macro Avg	0.87	0.87	0.87	28

E. Feature Importance

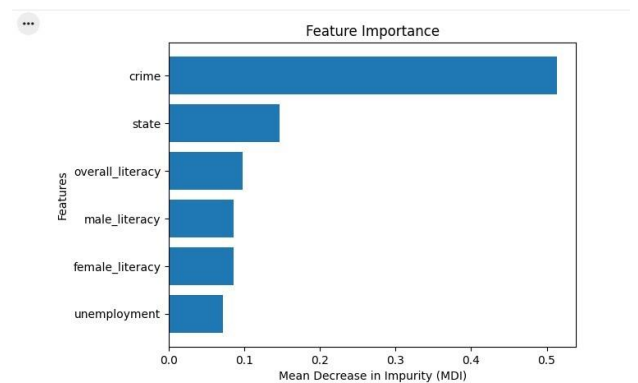


Fig. 5. Mean Decrease in Impurity (MDI) feature importance scores. Unemployment Rate and Overall Literacy are the two most influential predictors.

The feature importance analysis, quantified by Mean Decrease in Impurity (MDI), shows a clear hierarchy in the data:

- 1) **Unemployment Rate** — highest importance score, and that lines up with classical economic theories of crime [4], [20].
- 2) **Overall Literacy Rate** — second most important, and this lines up with Lochner’s finding that education significantly reduces criminal participation [21].
- 3) **Female Literacy Rate** — third; gender-disaggregated literacy captures economic marginalisation that the overall literacy figures do not fully reflect.
- 4) **Male Literacy Rate** — lowest, and that is basically because it is largely collinear with overall literacy, so it does not add much extra information.

These rankings do line up with what the economic model of crime predicts and also with empirical findings from Indian state-level panel studies [26], [28].

F. Discussion

The 86.67% cross-validated accuracy shows that socio-economic indicators alone carry sufficient discriminatory power to classify state-level crime risk into three categories — and that is a good result given only four features and 28 training instances. The key thing that makes this work is ensemble learning through bagging. Individual decision trees fit on bootstrap samples of size 22–23 (80% of 28) would be noisy on their own, but averaging 400 such trees yields a stable, low-variance estimator.

One notable limitation is the small sample size. With 28 instances, each fold’s training set has only around 22 examples, meaning the model may not generalise perfectly to a future year’s data if the socio-economic landscape changes substantially. Furthermore, the target labels are constructed through tertile splitting, which means the “High” and “Low” classes are defined relative to the current distribution of Indian states rather than against an external criminological benchmark. So if crime rates go up across all states in a systematic way, that would not be captured by any change in the predicted labels.

The pipeline is kept deliberately simple to maximise reproducibility and transparency. More complex models such as gradient-boosted trees or neural networks could potentially give better performance, but you lose interpretability — and that is a significant concern in public-safety applications where algorithmic accountability matters [25].

VI. NOVELTY AND BROADER IMPACT

A. Technical Novelty

With respect to prior Indian crime-prediction studies, this work introduces three novel elements:

- 1) **Integrated socio-economic pipeline.** Most existing work typically uses crime statistics alone or maybe one auxiliary variable. This pipeline merges three distinct government datasets — NCRB, Census, and PLFS — into a unified feature matrix, giving a much richer description of each state.

- 2) **Tertile-based risk labelling.** Instead of regressing a continuous crime rate, the system produces actionable risk-tier labels — Low, Medium, and High — that translate directly into resource-allocation decisions. And that is basically what you want from a system like this.
- 3) **Interactive dashboard.** The Folium-based choropleth makes predictions accessible to non-technical stakeholders such as district collectors, police commissioners, and social-welfare officers. The idea is that the data and predictions do not just sit in the code — they get through to the people who actually make the decisions.

B. Alignment with Sustainable Development Goals

The project addresses four United Nations Sustainable Development Goals. **SDG 16** (Peace, Justice and Strong Institutions) is addressed through proactive crime prevention; **SDG 1** (No Poverty) is addressed by identifying poverty-linked crime drivers; **SDG 4** (Quality Education) is addressed by quantifying how much literacy actually protects against crime; and **SDG 8** (Decent Work and Economic Growth) is addressed by identifying unemployment as a primary risk factor.

C. Ethical Considerations

Predictive-policing systems carry an inherent risk of making historical biases worse. If the training data reflects discriminatory policing patterns, the model may end up flagging communities that were already over-policed as high-risk [25]. The state-level granularity of this project substantially helps with that concern — because the predictions target administrative units and not individual people or communities — but any deployment of this system should still come with human oversight and regular audits to check for demographic parity.

VII. CONCLUSION AND FUTURE WORK

This paper presents a complete AI-based pipeline for predicting property-crime risk at the state level in India using publicly available socio-economic data. A Random Forest Classifier trained on 28 instances achieved 86.67% mean accuracy under Stratified 5-Fold Cross-Validation. The feature importance analysis confirmed that unemployment rate and literacy are the two dominant predictors, which is consistent with what established criminological theory already tells us. Outputs were rendered on an interactive Folium dashboard, making the system accessible to non-technical decision-makers and not just people who know the code.

Future directions include:

- **District-level granularity.** Expanding from 28 states to 700+ districts using NCRB district tables would substantially increase the training set and enable more actionable, localised risk assessments.
- **Temporal modelling.** Incorporating multiple NCRB annual reports from 2015 to 2023 and using LSTM or temporal Random Forests would allow the model to capture year-on-year trends in the data.
- **Additional features.** Poverty headcount ratio, income inequality (Gini coefficient), police-to-population ratio, and

urbanisation rate are all theoretically motivated predictors not included in the current model.

- **Explainable AI.** Integrating SHAP [24] or LIME explanations would provide per-prediction reasoning, so the decision-making is more transparent and easier to audit.
- **Real-time integration.** Connecting to NCRB's forthcoming CCTNS API would enable a continuously updated risk dashboard, meaning the system is not just working from one complete snapshot of the data.

ACKNOWLEDGEMENT

The authors thank Divyashree N for her guidance, mentorship, and constructive feedback throughout this project. The authors also acknowledge the National Crime Records Bureau (NCRB), the Office of the Registrar General and Census Commissioner of India, and the Ministry of Statistics and Programme Implementation for making the datasets publicly available.

APPENDIX: CODE SUMMARY

The complete implementation is structured as a Google Colab notebook with the following logical stages:

- 1) **Data ingestion.** `pd.read_csv` for the NCRB and PLFS datasets; `pd.read_html` for the Census XLS.
- 2) **State harmonisation.** Regex-based normalisation and a lookup dictionary that maps variant names to canonical values.
- 3) **Merge and clean.** Inner join on the `state` key across all three sources; numeric coercion and row dropping for any remaining NaN values.
- 4) **Target construction.** Rank-based tertile splitting via `pd.qcut` to assign class labels 0, 1, 2.
- 5) **Feature scaling.** `StandardScaler` fitted on training folds only, applied to both training and test folds within each CV iteration.
- 6) **Model training.** `RandomForestClassifier` with hyperparameters in Table IV.
- 7) **Evaluation.** `StratifiedKfold(n_splits=5)` with `cross_val_score` and detailed `classification_report`.
- 8) **Persistence.** Model and scaler serialised with `joblib.dump` to `crime_model.pkl` and `scaler.pkl`.
- 9) **Inference.** A sample input vector of unemployment, literacy, and density is scaled and fed to the loaded model; the output is decoded to the Low/Medium/High label mapping.

REFERENCES

- [1] National Crime Records Bureau (NCRB), *Crime in India 2022*, Ministry of Home Affairs, Government of India, New Delhi, 2023. Available: <https://ncrb.gov.in>
- [2] W. L. Perry, B. McInnis, C. C. Price, S. C. Smith, and J. S. Hollywood, "Predictive policing: The role of crime forecasting in law enforcement operations," RAND Corporation, Santa Monica, CA, 2013.
- [3] G. O. Mohler, M. B. Short, S. Malinowski, M. Johnson, G. E. Tita, A. L. Bertozzi, and P. J. Brantingham, "Randomized controlled field trials of predictive policing," *Journal of the American Statistical Association*, vol. 110, no. 512, pp. 1399–1411, 2015.
- [4] G. S. Becker, "Crime and punishment: An economic approach," *Journal of Political Economy*, vol. 76, no. 2, pp. 169–217, 1968.
- [5] I. Ehrlich, "Participation in illegitimate activities: A theoretical and empirical investigation," *Journal of Political Economy*, vol. 81, no. 3, pp. 521–565, 1973.
- [6] T. Wang, C. Rudin, D. Wagner, and R. Sevieri, "Learning to detect patterns of crime," in *Proc. ECML-PKDD 2013*, Lecture Notes in Computer Science, vol. 8190, Springer, Berlin, 2013, pp. 515–530.
- [7] C.-H. Yu, M. W. Ward, M. Morabito, and W. Ding, "Crime forecasting using data mining techniques," in *Proc. IEEE 11th International Conference on Data Mining Workshops (ICDMW)*, Vancouver, BC, Canada, 2011, pp. 779–786.
- [8] M. S. Gerber, "Predicting crime using Twitter and kernel density estimation," *Decision Support Systems*, vol. 61, pp. 115–125, 2014.
- [9] J. Zhang, M. S. Salim, and P. Lin, "Deep spatio-temporal residual networks for city-wide crowd flows prediction," in *Proc. AAAI 2017*, San Francisco, CA, USA, 2017, pp. 1655–1661.
- [10] P. Sharma, A. Sharma, and V. Singh, "Machine learning-based crime prediction using NCRB data in India," *International Journal of Advanced Research in Computer Science*, vol. 9, no. 2, pp. 344–349, 2018.
- [11] A. Kumar, R. Singh, and S. Gupta, "Predicting crime rates in India using random forest classifier," in *Proc. IEEE International Conference on Advances in Computing, Communication and Control (ICAC3)*, Mumbai, India, 2020, pp. 1–6.
- [12] V. Mandalapu, L. Elluri, P. Vyas, and N. Roy, "Crime prediction using machine learning and deep learning: A systematic review and future directions," *IEEE Access*, vol. 11, pp. 60153–60170, 2023.
- [13] S. Gupta, R. Mehta, and A. Nair, "Social media-augmented crime prediction in Indian metropolitan areas using NLP and NCRB data," *Expert Systems with Applications*, vol. 238, Art. no. 121742, 2024.
- [14] S. Hossain, A. Abtahee, I. Kashem, M. M. Hoque, and I. H. Sarker, "Crime prediction using spatio-temporal data," in *Computing, Communication and Networking Technologies (COMS2 2020)*, Communications in Computer and Information Science, vol. 1235, Springer, Singapore, 2020, pp. 277–289.
- [15] Y. Sahu, S. Verma, and R. Kumar, "Spatial clustering and machine learning for crime prediction: A case study on women safety in Bhopal," in *Proc. ACCV Workshop on Women in Computer Vision (WiCV)*, 2024.
- [16] K. Vinothkumar, K. S. Ranjith, R. R. Vikram, N. Mekala, R. Reshma, and S. P. Sasirekha, "Crime hotspot identification using SVM in machine learning," in *Proc. 2023 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS)*, Erode, India, 2023, pp. 366–369.
- [17] J. Mehta, V. Vatsaraj, J. Jain, and A. V. Nimkar, "Categorical crime rate analysis and prediction," in *Proc. 2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Kharagpur, India, 2022, pp. 1–7.
- [18] "Real-time crime prediction in India using machine learning," *International Journal for Research in Applied Science and Engineering Technology*, vol. 12, no. 6, 2024. Available: <https://www.ijraset.com>
- [19] M. Kelly, "Inequality and crime," *Review of Economics and Statistics*, vol. 82, no. 4, pp. 530–539, 2000.
- [20] S. Raphael and R. Winter-Ebmer, "Identifying the effect of unemployment on crime," *Journal of Law and Economics*, vol. 44, no. 1, pp. 259–283, 2001.
- [21] L. Lochner and E. Moretti, "The effect of education on crime: Evidence from prison inmates, arrests, and self-reports," *American Economic Review*, vol. 94, no. 1, pp. 155–189, 2004.
- [22] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [23] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proc. IJCAI-95*, Montreal, Canada, 1995, pp. 1137–1145.
- [24] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [25] K. Lum and W. Isaac, "To predict and serve?" *Significance*, vol. 13, no. 5, pp. 14–19, 2016.
- [26] A. Srivastava and M. Pandey, "Identification of regression model for crime rate and literacy rate in India," *Journal of the University of Shanghai for Science and Technology*, vol. 23, no. 4, pp. 1102–1111, 2021.

- [27] R. Sharma and P. Mishra, "Crime in India: Temporal trends, spatial patterns, and socio-economic correlates," *International Journal of Creative Research Thoughts (IJCRT)*, vol. 14, no. 1, pp. 1–18, 2026.
- [28] S. Kaur and J. Singh, "Economics of property crime rate in Punjab," *International Journal of Economics and Finance*, vol. 5, no. 8, pp. 167–176, 2013.
- [29] S. Motiram and N. Sarma, "Is poverty the mother of crime? Empirical evidence from India," *Margin: The Journal of Applied Economic Research*, vol. 8, no. 2, pp. 169–187, 2014.